期末考總複習

Orthogonal

Orthogonal = Independent?

- Any orthogonal set of nonzero vectors is linearly independent.

Let $\mathcal{S} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ be an orthogonal set $\mathbf{v}_i \neq \mathbf{0}$ for $i = 1, 2, \ldots, k$.

Assume $c_1, c_2, \ldots, c_k$ make $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$

$$
\begin{aligned}
&(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_i\mathbf{v}_i + \cdots + c_k\mathbf{v}_k) \cdot \mathbf{v}_i \\
&= \quad c_1\mathbf{v}_1 \cdot \mathbf{v}_i + c_2\mathbf{v}_2 \cdot \mathbf{v}_i + \cdots + c_i\mathbf{v}_i \cdot \mathbf{v}_i + \cdots + c_k\mathbf{v}_k \cdot \mathbf{v}_i \\
&= \quad c_i(\mathbf{v}_i \cdot \mathbf{v}_i) \quad = \quad c_i \underbrace{||\mathbf{v}_i||^2}_{\neq 0} \Rightarrow c_i = 0
\end{aligned}
$$

$\Rightarrow c_1 = c_2 = \cdots = c_k = 0$

https://speech.ee.ntu.edu.tw/~hylee/la/2021_materials/temp_materials/orthogonal%20basis%20(v2).pdf#page=4

Orthogonal Basis

- Let $S = \{v_1, v_2, \cdots, v_k\}$ be an orthogonal basis for a subspace W, and let u be a vector in W.

$$u = \underset{\displaystyle\frac{u \cdot v_1}{\|v_1\|^2}}{\underset{\downarrow}{c_1}} v_1 + \underset{\displaystyle\frac{u \cdot v_2}{\|v_2\|^2}}{\underset{\downarrow}{c_2}} v_2 + \cdots + \underset{\displaystyle\frac{u \cdot v_k}{\|v_k\|^2}}{\underset{\downarrow}{c_k}} v_k$$

- Let u be any vector, and w is the orthogonal projection of u on W.

$$w = \underset{\displaystyle\frac{u \cdot v_1}{\|v_1\|^2}}{\underset{\downarrow}{c_1}} v_1 + \underset{\displaystyle\frac{u \cdot v_2}{\|v_2\|^2}}{\underset{\downarrow}{c_2}} v_2 + \cdots + \underset{\displaystyle\frac{u \cdot v_k}{\|v_k\|^2}}{\underset{\downarrow}{c_k}} v_k$$

https://speech.ee.ntu.edu.tw/~hylee/la/2021_materials/temp_materials/orthogonal%20projection%20(v2).pdf#page=23

Orthogonal Projection

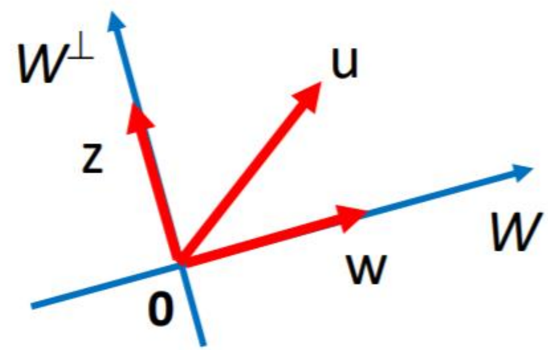
For any subspace W of R^n $\quad dimW + dimW^{\perp} = n$

$$\underbrace{\text{Basis: } \{w_1, w_2, \cdots, w_k\} \quad \text{Basis: } \{z_1, z_2, \cdots, z_{n-k}\}}_{\text{Basis for } R^n}$$

For ***every*** vector u,

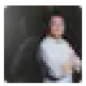
u = w + z ***(unique)***

$\in W$ $\qquad \in W^{\perp}$



https://speech.ee.ntu.edu.tw/~hylee/la/2021_materials/temp_materials/orthogonal%20projection%20(v2).pdf#page=10

Orthogonal Projection Matrix P_W



Let W be a subspace of R^n having dimension k, where 0 < k < n. P_W is the orthogonal projection matrix for W. Please find the eigenvalues of P_W.

The orthogonal projection matrix P_W has the following properties:

1. P_W is a symmetric matrix, i.e. P_W = P_W^T
2. P_W is a idempotent matrix, i.e. P_W^2 = P_W
3. P_W is a projection matrix, i.e. P_W x = x for all x in W

From these properties, it follows that the eigenvalues of P_W are either 0 or 1. The number of eigenvalues equal to 1 is equal to the dimension of W, k, and the number of eigenvalues equal to 0 is equal to n-k.

Therefore, the eigenvalues of P_W are 1, 1, ..., 1 (k times) and 0, 0, ..., 0 (n-k times).

Orthogonal Projection Matrix P_W

6. (6%) Let W be a subspace of $\mathcal{R}^n$ having dimension k, where $0 < k < n$. P_W is the orthogonal projection matrix for W. Please find the eigenvalues of P_W. (2018)

$$\begin{tikzcd} & \lambda = 1 \arrow[r] & W \\ \substack{P_W \\ k < n} \arrow[ur] \arrow[dr] & & \\ & \lambda = 0 \arrow[r] & W^{\perp} \end{tikzcd}$$

$$\substack{P_W \\ k = n} \longrightarrow \lambda = 1$$

(exercises 67, 72, 73, P464)

Orthogonal Projection Matrix P_W

2. Let W be a subspace of R^n. Prove that $(P_W)^2 = P_W$. (8%)

(2014)

5. Let W be a subspace of $\mathcal{R}^n$, and P_W is the orthogonal projection for W.
   (a) (5%) Prove that $P_W P_{W^\perp} = O$ (zero matrix).
   (b) (5%) Prove that $P_W + P_{W^\perp} = I_n$ (identity matrix).

(2018)

For ***every*** vector **u**,

$$\mathbf{u} = \underbrace{\mathbf{w}}_{\in W} + \underbrace{\mathbf{z}}_{\in W^\perp} \quad \textit{(unique)}$$

$$(P_W + P_{W^\perp})\boldsymbol{u}$$

$$= P_W\boldsymbol{u} + P_{W^\perp}\boldsymbol{u}$$

$$= \boldsymbol{w} + \boldsymbol{z} \quad = \boldsymbol{u}$$

(exercises 67, 72, 73, P464)

Orthogonal Projection Matrix P_W

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(2014)

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(a) (5%) Prove that $P_W P_{W^\perp} = O$ (zero matrix).
(b) (5%) Prove that $P_W + P_{W^\perp} = I_n$ (identity matrix).

(2018)

$$P_W = \frac{1}{6}\begin{bmatrix} 5 & 1 & -2 \\ 1 & 5 & 2 \\ -2 & 2 & 2 \end{bmatrix}$$

What is $\mathrm{P}_{W^\perp}$?

Orthogonal Matrix

Highlight

https://speech.ee.ntu.edu.tw/~tlkagk/courses/LA_2021/pdf/special%20matrix.pdf#page=7

- Q is an orthogonal matrix
- $QQ^T = I_n$
- Q is invertible, and $Q^{-1} = Q^T$ — Simple inverse
- $Qu \cdot Qv = u \cdot v$ for any u and v — Q preserves dot projects
- $\|Qu\| = \|u\|$ for any u — Q preserves norms

(c) (5%) Suppose

(2016) $$A = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

Find the inverse of matrix A. (Hint: Is A an orthogonal matrix?)

1. (5%) Let T be a linear operator on R^3 defined by $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} -x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Prove that T is an orthogonal operator. (2011)

5. The $n \times n$ matrix defined below is called a Householder matrix

$$P_v = I_n - 2\mathbf{v}\mathbf{v}^T,$$

where $\mathbf{v}$ is a unit vector in $\mathcal{R}^n$ (i.e., $\| \mathbf{v} \| = 1$). (2011)

(a) (5%) Prove that P_v is symmetric and orthogonal.

$$\|P_v u\| = \|u\|?$$

$$P_v u = (I - 2vv^T)u = u - 2vv^T u = u - 2v(v \cdot u) = u - 2(v \cdot u)v$$

$$\|P_v u\|^2 = (u - 2(v \cdot u)v) \cdot (u - 2(v \cdot u)v)$$

$$= \|u\|^2 - 4(v \cdot u)(v \cdot u) + 4(v \cdot u)(v \cdot u)\|v\|^2$$

$$= \|u\|^2 - 4(v \cdot u)(v \cdot u) + 4(v \cdot u)(v \cdot u) = \|u\|^2$$

1. (5%) Let T be a linear operator on R^3 defined by $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} -x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Prove that T is an orthogonal operator. (2011)

5. The $n \times n$ matrix defined below is called a Householder matrix

$$P_v = I_n - 2\mathbf{v}\mathbf{v}^T,$$

where $\mathbf{v}$ is a unit vector in $\mathcal{R}^n$ (i.e., $\| \mathbf{v} \| = 1$). (2011)

(a) (5%) Prove that P_v is symmetric and orthogonal.

(d) Determine whether the matrix A is orthogonal or not, where

$$A = I - 2uu^T$$

$$u = \frac{v}{\sqrt{v^T v}} \quad \|u\| = 1 \qquad A = I_n - \frac{2\mathbf{v}\mathbf{v}^T}{\mathbf{v}^T\mathbf{v}},$$

$\mathbf{v} \in R^n$ and I_n is an $n \times n$ identity matrix. (3%)
Justify your answer. (3%) (2014)

Symmetric Matrix

Symmetric matrix is diagonalizable

$A = PDP^{-1} = PDP^{T}$

$P = [u_1 \dots u_n]$ is orthogonal matrix

$\{u_1 \dots u_n\}$ is orthonormal basis of R^n

3. A is a 4×4 symmetric matrix.

(a) (5%) Is A diagonalizable? (Please explain your answer briefly.)

(2016)

(b) (5%) If all the eigenvalues of A are 2, please determine A.

$$
\begin{aligned}
A &= PDP^{-1} \\
&= P(2I)P^{-1} = 2PIP^{-1} \\
&= 2PP^{-1} = 2I
\end{aligned}
$$

$$
D = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix} = 2I
$$

(a) Let A be an $m \times n$ matrix. Please answer whether $A^T A$ is diagonalizable or not. (3%) Justify your answer. (3%)

(2014)

Because A^TA is symmetric, it is diagonalizable.

A is a 4x4 symmetric matrix. Is A diagonalizatble?

A symmetric matrix is diagonalizable if and only if all of its eigenvalues are distinct. To determine whether a 4x4 symmetric matrix is diagonalizable, you can find the eigenvalues and check if they are all distinct.

To find the eigenvalues of a matrix, you can use the characteristic equation, which is obtained by setting the determinant of the matrix equal to zero. For a 4x4 matrix, the characteristic equation is given by:

$|A - \lambda I| = 0$

where λ is the identity eigenvalue and I is the identity matrix.

You can then solve this equation for λ to find the eigenvalues of the matrix. If all of the eigenvalues are distinct, the matrix is diagonalizable. If some of the eigenvalues are repeated, the matrix may or may not b Try again ble, depending on the algebraic multiplicity (i.e., the number of times an eigenvalue is repeated) of the eigenvalues.

A is a 4x4 matrix. Is A diagonalizatble?

To determine whether a matrix is diagonalizable, we need to check if it is possible to find a diagonal matrix D and an invertible matrix P such that A = PDP^(-1).

In other words, we need to check if A can be written as a similarity transformation of a diagonal matrix.

To do this, we need to find the eigenvalues and eigenvectors of the matrix A. If the matrix A has n linearly independent eigenvectors, then it is diagonalizable. Otherwise, it is not diagonalizable.

To find the eigenvalues and eigenvectors of a matrix A, we need to solve the characteristic equation:

$|A - \lambda I| = 0$

Try again

where λ is an eigenvalue and I is the id

Spectral Decomposition

$A = PDP^{-1} = PDP^T$

$P = [u_1 \dots u_n]$ is orthogonal matrix

$\{u_1 \dots u_n\}$ is orthonormal basis of R^n

- Ref: https://youtu.be/0ijUQ-RfN3I?t=1880

7. Let A be an $n \times n$ symmetric matrix with spectral decomposition $A = \lambda_1 P_1 + \lambda_2 P_2 + \dots + \lambda_n P_n$, where λ_j's are distinct eigenvalues and $P_j = \mathbf{u}_j \mathbf{u}_j^T$, $\mathbf{u}_j$ is the corresponding normalized eigenvector of λ_j for all j.
(a) (4%) What is the spectral decomposition of A^2?
(b) (5%) Prove that $P_1 + P_2 + \dots + P_n = I_n$ (identity matrix).
(c) (5%) Prove that if the characteristic polynomial of A is $f(t) = a_n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0$, then $f(A) = O$ (zero matrix), where $f(A) = a_n A^n + a_{n-1} A^{n-1} + \dots + a_1 A + a_0 I_n$. Hint: There are two different approaches to this proof. You can use $A = PDP^{-1}$, where D is a diagonal matrix with eigenvalues as diagonal entries, or you can use $A = \lambda_1 P_1 + \lambda_2 P_2 + \dots + \lambda_n P_n$ and the fact of (b).

(2018)

$$A^2 = (\lambda_1)^2 P_1 + (\lambda_2)^2 P_2 + \dots\dots + (\lambda_n)^2 P_n$$

Exercise 53 (P500)

Spectral Decomposition

$A = PDP^{-1} = PDP^T$

$P = [u_1 \dots u_n]$ is orthogonal matrix

$\{u_1 \dots u_n\}$ is orthonormal basis of R^n

(c) Exercise 56 (P500), Exercise 88 (P323)

- Ref: https://youtu.be/0ijUQ-RfN3I?t=1880

7. Let A be an $n \times n$ symmetric matrix with spectral decomposition $A = \lambda_1 P_1 + \lambda_2 P_2 + \dots + \lambda_n P_n$, where λ_j's are distinct eigenvalues and $P_j = \mathbf{u}_j \mathbf{u}_j^T$, $\mathbf{u}_j$ is the corresponding normalized eigenvector of λ_j for all j.
(a) (4%) What is the spectral decomposition of A^2?
(b) (5%) Prove that $P_1 + P_2 + \dots + P_n = I_n$ (identity matrix).
(c) (5%) Prove that if the characteristic polynomial of A is $f(t) = a_n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0$, then $f(A) = O$ (zero matrix), where $f(A) = a_n A^n + a_{n-1} A^{n-1} + \dots + a_1 A + a_0 I_n$. Hint: There are two different approaches to this proof. You can use $A = PDP^{-1}$, where D is a diagonal matrix with eigenvalues as diagonal entries, or you can use $A = \lambda_1 P_1 + \lambda_2 P_2 + \dots + \lambda_n P_n$ and the fact of (b).

(2018)

Show that $(P_1 + P_2 + \cdots + P_n)v = v$

$\{u_1 \dots u_n\}$ is orthonormal basis of R^n

$v = c_1 u_1 + c_2 u_2 + \cdots + c_n u_n$

Vector Space

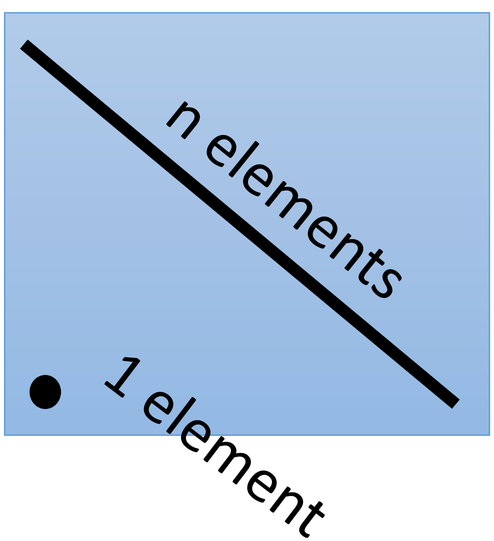
Dimension

8. (a) (4%) Let V be the vector space of all $n \times n$ matrices with real entries. Let W be the set of all $n \times n$ symmetric matrices with real entries. Prove that W is a subspace of V.

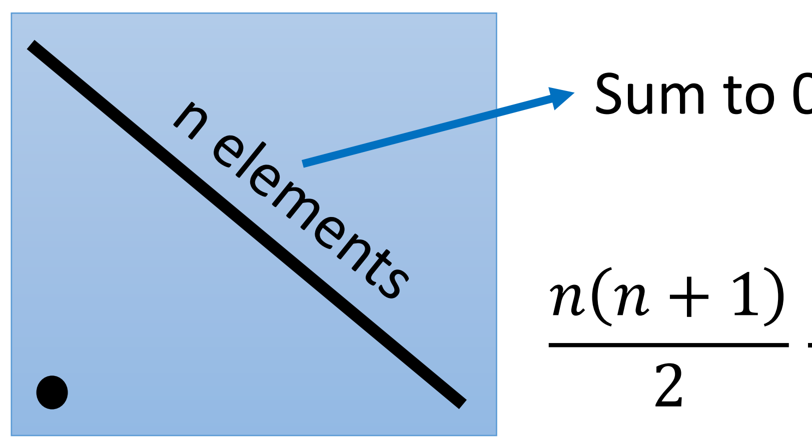
(2018)

(b) (3%) What is the dimension of W? No proof is needed.

(c) (3%) Let Z be the subspace of W such that for each matrix in Z, the sum of diagonal entries is zero. What is the dimension of Z? No proof is needed.



$$\frac{n(n+1)}{2}$$



$$\frac{n(n+1)}{2} - 1$$

Dimension

Exercise 76, P392:

$$dimL(V,W) = dimV \cdot dimW$$

4. Let $\mathscr{L}(M_{n\times n}, M_{n\times n})$ be the set of all linear transformations from $M_{n\times n}$ to $M_{n\times n}$. Let $T, U \in \mathscr{L}(M_{n\times n}, M_{n\times n})$ and $c \in R$.
Define $T + U : M_{n\times n} \to M_{n\times n}$ and $cT : M_{n\times n} \to M_{n\times n}$ by

(2013)

$$(T+U)(A) = T(A) + U(A)$$
$$(cT)(A) = c\,T(A) \quad \text{for all } A \in M_{n\times n}$$

Thus, $V = \mathscr{L}(M_{n\times n}, M_{n\times n})$ is a vector space over R.
What is the dimension of V? No proof is needed. (5%)

n x n x n x n

6. Let $M_{m\times n}$ be the set of all real $m \times n$ matrices, which is a vector space.

   (a) (5%) Let $S_{n\times n}$ be the set of all real symmetric $n \times n$ matrices. Prove that $S_{n\times n}$ is a subspace of $M_{n\times n}$.

   (b) (5%) Let $\mathcal{L}(M_{m\times n}, M_{m\times n})$ be the set of all linear transformations from $M_{m\times n}$ to $M_{m\times n}$, which is a vector space. Please find the dimension of $\mathcal{L}(M_{m\times n}, M_{m\times n})$. No derivation is needed.

m x n x m x n (2011)

(c) (5%) Prove that every square matrix can be written uniquely as the sum of a symmetric matrix and a skew-symmetric matrix. That is, if C is an $n \times n$ matrix, then $C = B + A$, where B and A are uniquely determined by C such that $B^T = B$ and $A^T = -A$.

(2012)

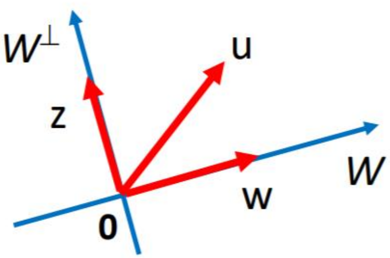
For any subspace W of R^n $\quad dimW + dimW^{\perp} = n$

$\underbrace{\text{Basis: } \{w_1, w_2, \cdots, w_k\} \quad \text{Basis: } \{z_1, z_2, \cdots, z_{n-k}\}}_{\text{Basis for } R^n}$

For ***every*** vector u,

u = w + z ***(unique)***

$\in W$ $\quad \in W^{\perp}$



W: symmetric matrix

$W^{\perp}$: skew-symmetric matrix

$$C = B + A$$

$$B \in W, A \in W^{\perp}$$

$$B = \frac{C + C^T}{2} \quad A = \frac{C - C^T}{2}$$

8. Consider the vector space of 2×2 real matrices. The inner product of A, B is defined as $< A, B >= a_{11}b_{11} + a_{12}b_{12} + a_{21}b_{21} + a_{22}b_{22}$. Find the 2×2 real symmetric matrix that is closest to $\begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}$. (10%)

(2013)

Matrix representation of linear operator

(Section 4.5, 6.4)

9. Let $T : \mathcal{P}_n \to \mathcal{P}_n$ be defined by $T(\sum_{i=0}^n a_i x^i) = \sum_{i=0}^n a_{n-i} x^i$. (2018)
(a) (4%) Prove that T is a linear transformation.
(b) (4%) Define the inner product for P_n by $\langle f(x), g(x)\rangle = \sum_{i=0}^n f_i g_i$ for $f(x) = \sum_{i=0}^n f_i x^i$ and $g(x) = \sum_{i=0}^n g_i x^i$. Prove that $[T]_\beta$ is an orthogonal matrix for any orthonormal basis β of P_n.

---

8. Let V the set of all 2×2 symmetric matrices. Let $T : V \to V$ be the linear transformation defined by $T(A) = AB + BA$, where $B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.

(2016)

(a) (6%) Find a basis $\mathcal{B}$ for V.

(b) (8%) Find $[T]_\mathcal{B}$, the matrix representation of T with respective to the basis $\mathcal{B}$ in Part (a).

(c) (6%) Find a basis for the range of T.

1. (15%) Let $T : P_2 \to P_2$ be a linear operator defined by $T(a + bx + cx^2) = (3a + c) + (a + b)x + (-a - b + 3c)x^2$. Let $\mathcal{B}_1 = \{1, x, x^2\}$ and $\mathcal{B}_2 = \{1 + x + x^2, 1 + 2x + 3x^2, 2 + x + x^2\}$ be two bases for P_2.

(a) (7%) Find $T_{\mathcal{B}_1}$, the matrix representation of T with respect to $\mathcal{B}_1$.

(b) (8%) Find $T_{\mathcal{B}_2}$, the matrix representation of T with respect to $\mathcal{B}_2$. (2015)

Matrix representation of linear operator

(Section 4.5, 6.4)

4. (20%) A matrix is said to be skew-symmetric if $A^T = -A$. Let W be the set of all 3×3 skew-symmetric matrices. Let $T : W \to W$ be defined by $T(A) = CAC^T$, where

$$C = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}.$$

(2015)

(a) (8%) Find a basis $\mathcal{B}$ for W.

(b) (6%) Find $[T]_{\mathcal{B}}$, where $\mathcal{B}$ is the basis for W in Part (a).

(c) (6%) Find a basis for the null space of T.

7. Define a linear transformation T from P_2 to P_2 by $T(p(x)) = p(0) + 3p(1)x + p(2)x^2$. Let $\mathcal{B} = \{1, 1 + x, 1 + x + x^2\}$ be a basis for P_2.

(a) (7%) Let $p(x) = 2 + x - 2x^2$. Find the coordinate vector of $p(x)$ relative to $\mathcal{B}$.

(b) (8%) Find the matrix representation of T with respect to $\mathcal{B}$.

(c) (5%) Find the coordinate vector of $T(p(x))$ relative to $\mathcal{B}$.

(2011)

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